ClipSAT : Math for AI & Data Science

Unit 6.01: Linear Regression & OLS

Name: _____

Q1. A fitted simple linear regression model is hat{y} = 2x + 3 . What is the predicted target value hat{y} when the input feature is x = 4 ?

[ ] A: 11
[ ] B: 14
[ ] C: 7
[ ] D: 8

Q2. A fitted simple linear regression model is hat{y} = 3x + 6 . What is the predicted target value hat{y} when the input feature is x = 4 ?

[ ] A: 18
[ ] B: 21
[ ] C: 14
[ ] D: 12

Q3. A fitted simple linear regression model is hat{y} = 4x + 9 . What is the predicted target value hat{y} when the input feature is x = 4 ?

[ ] A: 25
[ ] B: 28
[ ] C: 21
[ ] D: 16

Q4. A fitted simple linear regression model is hat{y} = 5x + 12 . What is the predicted target value hat{y} when the input feature is x = 4 ?

[ ] A: 32
[ ] B: 35
[ ] C: 28
[ ] D: 20

Q5. A fitted simple linear regression model is hat{y} = 6x + 15 . What is the predicted target value hat{y} when the input feature is x = 4 ?

[ ] A: 39
[ ] B: 42
[ ] C: 35
[ ] D: 24

Q6. A fitted simple linear regression model is hat{y} = 7x + 18 . What is the predicted target value hat{y} when the input feature is x = 4 ?

[ ] A: 46
[ ] B: 49
[ ] C: 42
[ ] D: 28

Q7. In a multiple linear regression model, the Sum of Squared Errors is text{SSE} = 20 and the Total Sum of Squares is text{SST} = 100 . What is the coefficient of determination R^2 ?

[ ] A: R^2 = 0.80
[ ] B: R^2 = 0.20
[ ] C: R^2 = 0.90
[ ] D: R^2 = 0.25

Q8. In a multiple linear regression model, the Sum of Squared Errors is text{SSE} = 40 and the Total Sum of Squares is text{SST} = 200 . What is the coefficient of determination R^2 ?

[ ] A: R^2 = 0.80

[ ] B: R^2 = 0.20

[ ] C: R^2 = 0.90

[ ] D: R^2 = 0.25